

A1 Basic Linear Algebra

A1.1 Notation

We will use a bold capital letter to indicate a matrix. Each element of a matrix has two subscripts, x_{ij} , the first indicates the column and the second indicates the row, e.g.,

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix},$$

Vectors will also be bold, but we will use lower case letters instead to refer to them $\mathbf{x}$. MATLAB uses matrix notation for indices, which makes indices seem a little backwards for Cartesian coordinates where j is the x index and i is the y index.

A1.2 Special Matrices

There are two special matrices that we will use. The first is the identity matrix, $\mathbf{I}$, which is a diagonal matrix of ones and is the equivalent of multiplying a matrix by one. A matrix multiplied by the identity matrix is itself, i.e., $\mathbf{XI} = \mathbf{X}$. An example 3×3 identity matrix is

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

To create an identity matrix in MATLAB with dimensions $M \times M$,

`eye (M)`

The second special matrix is really a column vector of ones, $\mathbf{1}$, e.g.,

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

In MATLAB, you can easily create a column vector of ones with length M by

`ones ( [M, 1] )`

A1.3 Basic Operations

A.1.3.1 Matrix Addition

Scalar addition is performed by simply adding the scalar to each element of the matrix, e.g.,

$$\mathbf{X} + a = a + \mathbf{X} = \begin{pmatrix} x_{11} + a & x_{12} + a & x_{13} + a \\ x_{21} + a & x_{22} + a & x_{23} + a \end{pmatrix}$$

Subtraction is handled in the same way.

A.1.3.2 Scalar–Matrix Multiplication/Division

Scalar multiplication is preformed by simply multiplying the the scalar to each element of the matrix, e.g.,

$$a\mathbf{X} = \mathbf{X}a = \begin{pmatrix} ax_{11} & ax_{12} & ax_{13} \\ ax_{21} & ax_{22} & ax_{23} \end{pmatrix}$$

Division by a scalar

$$\mathbf{X}/a = \begin{pmatrix} x_{11}/a & x_{12}/a & x_{13}/a \\ x_{21}/a & x_{22}/a & x_{23}/a \end{pmatrix}$$

A.1.3.3 Vector–Matrix Multiplication

To multiply a matrix by a vector, the inner dimensions of the matrix and vector must be equal. That is an $M \times N$ matrix can be multiplied by a row matrix of length M on the left or by a column matrix of length N on the right. The product will be a vector with the length of the matrix outer dimension in the same orientation of the multiplying vector. For a row matrix, the result will be $1 \times N$ e.g.,

$$\mathbf{u}_{1 \times 2} = (u_{11} \quad u_{12})$$

$$\mathbf{X}_{2 \times 3} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}$$

$$\mathbf{u}\mathbf{X} = (u_{11} \quad u_{12}) \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix} = (u_{11}x_{11} + u_{12}x_{21} \quad u_{11}x_{12} + u_{12}x_{22} \quad u_{11}x_{13} + u_{12}x_{23})$$

and for a column vector the result will be $M \times 1$, e.g.,

$$\mathbf{v}_{3 \times 1} = \begin{pmatrix} v_{11} \\ v_{21} \\ v_{31} \end{pmatrix}$$

$$\mathbf{X}\mathbf{v} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \\ v_{31} \end{pmatrix} = \begin{pmatrix} x_{11}v_{11} + x_{12}v_{21} + x_{13}v_{31} \\ x_{21}v_{11} + x_{22}v_{21} + x_{23}v_{31} \end{pmatrix}$$

Something interesting happens if one multiplies vectors. A column vector on the left can always be multiplied by a row vector on the right, resulting in a matrix, e.g.,

$$\mathbf{v}\mathbf{u} = \begin{pmatrix} v_{11} \\ v_{21} \\ v_{31} \end{pmatrix} (u_{11} \quad u_{12}) = \begin{pmatrix} v_{11}u_{11} & v_{11}u_{12} \\ v_{21}u_{11} & v_{21}u_{12} \\ v_{31}u_{11} & v_{31}u_{12} \end{pmatrix}$$

but a row vector on the left can only be multiplied by a column vector on the right when the lengths are equal, the result is a single value, e.g.,

$$\mathbf{w}_{1 \times 3} = (w_{11} \quad w_{12} \quad w_{13})$$

$$\mathbf{w}\mathbf{v} = \begin{pmatrix} w_{11} & w_{12} & w_{13} \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{21} \\ v_{31} \end{pmatrix} = w_{11}v_{11} + w_{12}v_{21} + w_{13}v_{31}$$

A.1.3.4 Matrix–Matrix Multiplication

To multiply two matrices, the inner dimensions of the matrices must be equal. That is an $M \times N$ matrix can be multiplied by a $P \times M$ matrix on the left, or a $N \times P$ matrix on the right. For example,

$$\mathbf{X}_{2 \times 3} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}$$

$$\mathbf{Y}_{3 \times 2} = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \\ y_{31} & y_{32} \end{pmatrix}$$

will result in an $M \times P$ for a right multiplication,

$$\mathbf{XY} = \begin{pmatrix} x_{11}y_{11} + x_{12}y_{21} + x_{13}y_{31} & x_{11}y_{12} + x_{12}y_{22} + x_{13}y_{32} \\ x_{21}y_{11} + x_{22}y_{21} + x_{23}y_{31} & x_{21}y_{12} + x_{22}y_{22} + x_{23}y_{32} \end{pmatrix}$$

and will result in a $P \times N$ for a left multiplication,

$$\mathbf{YX} = \begin{pmatrix} y_{11}x_{11} + y_{12}x_{21} & y_{11}x_{12} + y_{12}x_{22} & y_{11}x_{13} + y_{12}x_{23} \\ y_{21}x_{11} + y_{22}x_{21} & y_{21}x_{12} + y_{22}x_{22} & y_{21}x_{13} + y_{22}x_{23} \\ y_{31}x_{11} + y_{32}x_{21} & y_{31}x_{12} + y_{32}x_{22} & y_{31}x_{13} + y_{32}x_{23} \end{pmatrix}$$

In the case above, the multiplication order could be reversed to produce a valid, albeit different, result. In general, it is not possible to reverse the order because the dimensions may not be compatible for multiplication.

A1.4 Special Matrix Operations

A1.4.1 Determinant

A determinant is a scalar value determined from a square matrix that can be thought of as a signed volume. The generic formulation is

$$\det(\mathbf{X}) = \sum_{\sigma \in S_n} \left(\text{sgn}(\sigma) \prod_{i=1}^n x_{i,\sigma_i} \right)$$

where σ is drawn from the set of all permutations of the set $S_n = \{1, 2, \dots, n\}$. The sign of σ results in +1 or -1 can be determined by reordering σ by swapping adjacent values. When the number of swaps required to order the set is even, the sign is + and when the number of swaps is odd, the sign is -. Let's look at examples for 2x2 and 3x3 matrices. For a 2x2,

$$\det(\mathbf{X}) = \begin{vmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{vmatrix} = x_{11}x_{22} - x_{12}x_{21}.$$

Note that sometimes the vertical bars, #, are used to indicate a determinate. For a 3x3,

$$\det(\mathbf{X}) = \begin{vmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{vmatrix},$$

we start with producing the permutations of σ ,

σ	$\text{sgn}(\sigma)$
$\{1, 2, 3\}$	+1
$\{1, 3, 2\}$	-1
$\{2, 1, 3\}$	-1
$\{2, 3, 1\}$	+1
$\{3, 1, 2\}$	+1
$\{3, 2, 1\}$	-1

$$\det(\mathbf{X}) = \prod_{i=1}^3 a_{i,\{1,2,3\}} - \prod_{i=1}^3 a_{i,\{1,3,2\}} - \prod_{i=1}^3 a_{i,\{2,1,3\}} + \prod_{i=1}^3 a_{i,\{2,3,1\}} + \prod_{i=1}^3 a_{i,\{3,1,2\}} - \prod_{i=1}^3 a_{i,\{3,2,1\}}$$

$$\det(\mathbf{X}) = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

There are simpler and more visual methods for 3x3, but these do not extend to larger matrices. Thankfully, computers can handle this very quickly and easily so we don't have to.

A1.4.2 Inverse

An inverse of a matrix, when multiplied by the original matrix must result in the identity matrix, i.e.,

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$$

The inverse can be expressed as

$$\mathbf{X}^{-1} = \frac{1}{\det(\mathbf{X})} \mathbf{C}^T$$

where $\mathbf{C}$ is the cofactor matrix of $\mathbf{X}$, where $c_{ij} = (-1)^{i+j} \det(\mathbf{C}_{ji})$ and $\mathbf{C}_{ji}$ is $\mathbf{C}$ with the i -th row and j -th column removed. Because the inverse of a matrix depends upon the determinant, it can only be performed on a square matrix with a non-zero determinant.

A1.4.3 Transpose

We will commonly use a matrix transpose in our computations. The operator for transpose is a superscript T, T , although you may find some texts use ', which also happens to be the MATLAB operator for transpose, e.g.,

$$\mathbf{x}'$$

The operation is rather simple as it swaps the order of the subscripts, $x_{ij}^T = x_{ji}$. A transpose will turn a row vector into a column vector and for a full matrix,

$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}$, the dimensions will be swapped as the values are reflected across the diagonal,

$$\mathbf{X}^T = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \end{pmatrix}.$$

There are several important properties of a transpose:

1. $(\mathbf{X}^T)^T = \mathbf{X}$
2. $(\mathbf{X} + \mathbf{Y})^T = \mathbf{X}^T + \mathbf{Y}^T$, where $\mathbf{Y}$ is a matrix with the same dimensions as $\mathbf{X}$.
3. $(a\mathbf{X})^T = a\mathbf{X}^T$, where a is a scalar.
4. $(\mathbf{XY})^T = \mathbf{Y}^T\mathbf{X}^T$
5. $(\mathbf{X}^T)^{-1} = (\mathbf{X}^{-1})^T$

A1.5 Matrix Properties

A1.5.1 Symmetric Matrix

A matrix is symmetric if $\mathbf{X}^T = \mathbf{X}$, i.e., $x_{ij} = x_{ji}$ for all i and j . A symmetric matrix must be square. An example of a symmetric matrix is

$$\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$$

Note how the values are symmetric about the diagonal.

A1.5.2 Singular Matrix

A matrix is singular if any row is all zeros, any column is all zeros, two columns that are the same, or two rows that are the same, e.g.,

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$$

singular matrices must be square, they do not have an inverse and their determinant is 0. Note that matrices can also be singular if two rows (or columns) are equal after multiplying by a scalar, e.g.,

$$\begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}$$

is singular because the second column is negative twice that of the first column. For our data collection, this would be the equivalent of taking two identical sets of data at the same point.